WORKED SOLUTION

STEP III 1990, Question 1

Pure • Multiple Angles and Polynomial Roots

STEP III 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q1

The key idea

Nine is 3×3, so we can get $\tan 9\theta$ by using the triple-angle formula for tangent twice. This is much less work than expanding $(\cos\theta + \mathrm{i}\sin\theta)^9$. The factorised form also tells us which factor vanishes at $\theta = \pi/9, 2\pi/9, 4\pi/9$: these are exactly the angles where $\tan^2 3\theta = 3$. That gives a cubic in $u = \tan^2\theta$, and Vieta's formulas (the links between a polynomial's roots and its coefficients) do the rest.

Solution

The formula for $\tan 9\theta$

Write $t = \tan\theta$ and $T = \tan 3\theta$. The triple-angle formula, which you can get from $\tan(2\theta + \theta)$ or from de Moivre with $n = 3$, gives

$$T = \frac{3t - t^3}{1 - 3t^2} = \frac{t(t^2 - 3)}{3t^2 - 1}, \qquad \tan 9\theta = \frac{3T - T^3}{1 - 3T^2} = \frac{T(3 - T^2)}{1 - 3T^2}.$$

We now write $3 - T^2$ and $1 - 3T^2$ over the common denominator $(1 - 3t^2)^2$:

$$\begin{aligned}(1 - 3t^2)^2(3 - T^2) &= 3(1 - 3t^2)^2 - t^2(3 - t^2)^2 \\ &= (3 - 18t^2 + 27t^4) - (9t^2 - 6t^4 + t^6) \\ &= -(t^6 - 33t^4 + 27t^2 - 3),\end{aligned}$$

$$\begin{aligned}(1 - 3t^2)^2(1 - 3T^2) &= (1 - 3t^2)^2 - 3t^2(3 - t^2)^2 \\ &= (1 - 6t^2 + 9t^4) - (27t^2 - 18t^4 + 3t^6) \\ &= -(3t^6 - 27t^4 + 33t^2 - 1).\end{aligned}$$

The factors $-(1 - 3t^2)^{-2}$ cancel in the ratio, so

$$\tan 9\theta = \frac{t(t^2 - 3)}{3t^2 - 1} \cdot \frac{t^6 - 33t^4 + 27t^2 - 3}{3t^6 - 27t^4 + 33t^2 - 1},$$

which is the required result.

Using de Moivre instead, $\tan 9\theta = \operatorname{Im}(1+it)^9 / \operatorname{Re}(1+it)^9$. One then has to check that

$$9t - 84t^3 + 126t^5 - 36t^7 + t^9 \quad \text{and} \quad 1 - 36t^2 + 126t^4 - 84t^6 + 9t^8$$

factorise as the numerator and the denominator above. They do, but the route through $\tan 3\theta$ hands us the factorisation for free.

The cubic

Let $P(t) = t^6 - 33t^4 + 27t^2 - 3$. The first calculation above says

$$3 - \tan^2 3\theta = -\frac{P(t)}{(1-3t^2)^2} \qquad \text{whenever } 3t^2 \neq 1.$$

Now take $\theta = \pi/9,\ 2\pi/9,\ 4\pi/9$. Then $3\theta = \pi/3,\ 2\pi/3,\ 4\pi/3$, and in each case $\tan 3\theta = \pm\sqrt{3}$, so $\tan^2 3\theta = 3$. In each case 3θ is not an odd multiple of $\pi/6$, so $3t^2 \neq 1$ (otherwise $\tan 3\theta$ would be undefined). Hence $P(\tan\theta) = 0$ for all three angles.

These are exactly the angles in $(0, \pi/2)$ at which $\tan 9\theta = 0$ through the factor P rather than through t or $t^2 - 3$. Indeed, $t \neq 0$, and $t^2 \neq 3$ because $\theta \neq \pi/3$.

> **Ask yourself** How do we know that these three numbers are *all* the roots of the cubic, with none repeated?

Put $u = t^2$. Each of $\tan^2(\pi/9)$, $\tan^2(2\pi/9)$, $\tan^2(4\pi/9)$ is a root of

$$u^3 - 33u^2 + 27u - 3 = 0.$$

The three values are distinct because tan is positive and strictly increasing on $(0, \pi/2)$, and $\pi/9 < 2\pi/9 < 4\pi/9 < \pi/2$. A cubic has at most three roots, so these are precisely its roots. This is the required cubic with integer coefficients.

The product

By Vieta's formulas, the product of the roots of $u^3 - 33u^2 + 27u - 3$ is 3:

$$\tan^2\left(\tfrac{\pi}{9}\right)\tan^2\left(\tfrac{2\pi}{9}\right)\tan^2\left(\tfrac{4\pi}{9}\right) = 3.$$

All three angles lie in $(0, \pi/2)$, so each tangent is positive, and we take the positive square root:

$$\tan\left(\tfrac{\pi}{9}\right)\tan\left(\tfrac{2\pi}{9}\right)\tan\left(\tfrac{4\pi}{9}\right) = \sqrt{3}.$$

The sum of sixth powers

Let the roots be u_1, u_2, u_3. From the cubic,

$$e_1 = u_1 + u_2 + u_3 = 33, \qquad e_2 = u_1u_2 + u_2u_3 + u_3u_1 = 27, \qquad e_3 = u_1u_2u_3 = 3.$$

We want $u_1^3 + u_2^3 + u_3^3$, which is the sum of the sixth powers of the tangents.

Ask yourself How do we get a sum of cubes from e_1, e_2, e_3 without finding the roots?

Each root satisfies $u_i^3 = 33u_i^2 - 27u_i + 3$. Adding these three equations gives

$$\sum u_i^3 = 33\sum u_i^2 - 27\sum u_i + 9.$$

Here $\sum u_i = 33$, and

$$\sum u_i^2 = e_1^2 - 2e_2 = 1089 - 54 = 1035.$$

Therefore

$$\begin{aligned}\sum u_i^3 &= 33 \times 1035 - 27 \times 33 + 9\\ &= 34155 - 891 + 9\\ &= 33273.\end{aligned}$$

That is,

$$\tan^6\left(\tfrac{\pi}{9}\right) + \tan^6\left(\tfrac{2\pi}{9}\right) + \tan^6\left(\tfrac{4\pi}{9}\right) = 33273,$$

as required. Equivalently, $\sum u_i^3 = e_1^3 - 3e_1e_2 + 3e_3 = 35937 - 2673 + 9$.

Checks

- Numerically, $\tan^2(\pi/9) \approx 0.1325$, $\tan^2(2\pi/9) \approx 0.7041$ and $\tan^2(4\pi/9) \approx 32.163$. Their sum is 33.000, and their product is 3.000.
- The product of the three tangents is $1.7320508\ldots = \sqrt{3}$.
- The sum of the cubes of the three values above is 33273.0. Almost all of it comes from $\tan^6(4\pi/9) \approx 33272.4$.
- The formula for $\tan 9\theta$ agrees with a calculator to about twelve significant figures at thousands of random values of θ.
- Small case: put $t = 1$, so $\theta = \pi/4$. The formula gives $\frac{1\cdot(-2)(-8)}{2\cdot 8} = 1 = \tan(9\pi/4)$.

Common slips

- Forgetting to check that $3t^2 \neq 1$ (or that the denominator is non-zero) before concluding that the sextic factor vanishes.
- Taking $-\sqrt{3}$ for the product. Every angle is acute, so every tangent is positive.
- Writing $\sum u_i^3 = e_1^3 - 3e_1e_2$ without the $+3e_3$ term, which gives 33264.

Beyond STEP

The cubic $u^3 - 33u^2 + 27u - 3$ has no rational roots, so it is irreducible over $\mathbb{Q}$. It also has three real roots. Galois theory shows that the roots of such a cubic cannot be written

using real radicals alone (the *casus irreducibilis*). That is why trigonometric expressions like $\tan^2(\pi/9)$ are the natural way to write them down. Sums of powers of roots are governed in general by Newton's identities. For the geometric side of de Moivre's theorem, read Needham, *Visual Complex Analysis*. For the algebra, read Allenby, *Rings, Fields and Groups*.

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